

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1450

Imrie 100 kV
February 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1450. The Interconnection Customer proposes a 80.5 MW battery energy storage facility interconnecting at the Imrie 100 kV substation in the Michigan area. The request was received on April 09, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 412 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$46,239,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
New Hanover County Wastec - Sister Lakes 100 KV	Sister Lakes - Berrien Springs 100 KV	123.6	94	49.8
New Hanover County Wastec - Cantonment 100 KV	Sister Lakes Substation - Berrien Springs 100 KV	117.7	117	49.8
W Loop Tap Sta - El Sol 100 KV	Imperial Valley - Medway Hydro 100 KV	115.0	193	100.0
El Sol - Sister Lakes 100 KV	Imperial Vly - Medway Hydro 100 KV	112.7	213	100.0
Medway Hydro - Imperial Vly 100 KV	Sister Lakes Substation - El Sol 100 KV L	112.7	213	100.0
Berrien Springs Substation - N Mesquite 100 KV Line 1	N Mesquite Substation - Berrien Springs 100 KV Cir. 2	110.0	153	38.5
Berrien Springs - Sister Lakes Substation 100 KV	Imperial Valley - Medway Hydro 100 KV	108.1	177	64.7
Berrien Springs - North Mesquite 100 KV	North Mesquite - Berrien Springs 100 KV	106.6	158	38.5
Hagar - West Loop Tap Station 100 KV	Imperial Vly - Medway Hydro 100 KV	100.5	124	54.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
New Hanover County Wastec Substation - Sister Lakes 100 KV	Rebuild line with higher-capacity conductor	\$11,299,000	100% (\$11,299,000)
New Hanover County Wastec - Cantonment 100 KV	Rebuild line with higher-capacity conductor	\$2,828,000	100% (\$2,828,000)
W Loop Tap Sta - El Sol 100 KV	Rebuild line with higher-capacity conductor	\$4,431,000	100% (\$4,431,000)
El Sol - Sister Lakes 100 KV	Rebuild line with higher-capacity conductor	\$5,152,000	100% (\$5,152,000)
Medway Hydro - Imperial Vly 100 KV	Rebuild line with higher-capacity conductor	\$11,492,000	100% (\$11,492,000)
Berrien Springs Substation - N Mesquite 100 KV Line 1	Reconductor line	\$2,984,000	100% (\$2,984,000)
Berrien Springs - Sister Lakes Substation 100 KV	Reconductor line	\$3,165,000	100% (\$3,165,000)
Berrien Springs - N Mesquite 100 KV Line 2	Reconductor line	\$3,276,000	100% (\$3,276,000)
Hagar - West Loop Tap Station 100 KV	Reconductor line	\$1,612,000	100% (\$1,612,000)
	Total allocated to project		\$46,239,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.